STAT3003: Survey Methods
7. Beyond Formulas

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7.1 Errors of Non-observation

- Let us consider where our sample survey has errors
- We will classify errors into two groups
  - Errors of non-observation
  - Errors of observation
- Can you think of any error from sample surveys?

7.1 Errors of Non-observation

- Usually, the data we observe in a sample is not exactly the same as the data in the population
- This is an unavoidable consequence of conducting a sample survey instead of a census
- We call this sampling error
  - The difference between our estimate and the true parameter value due to the sampling

7.1 Errors of Non-observation

- How can we reduce sampling error?
  - Use an appropriate sample design
  - Use as large a sample size as possible
- All the designs and formulas for estimators and variance can be used to reduce this source of error
- The investigator has some control over it, through the methods we have learned so far

7.1 Errors of Non-observation

- What if our sampling frame does not contain every sampling unit in the population
  - E.g. phonebooks will not contain unlisted telephone numbers
  - E.g. lists can be outdated – things may have changed since they were drawn up
- This is called error of coverage
  - And it is difficult to measure or correct
- Reports should consider error of coverage

7.1 Errors of Non-observation

- Non-response is perhaps the most serious non-observation error
- Non-response can happen in one of the following ways
  - Inability to contact the sample element (e.g. person not at home)
  - Inability of sample element to answer the question (e.g. person doesn't know how much savings they have)
  - Refusal to answer question

7.1 Errors of Non-observation

- Inability to contact
  - We must make observations of exactly the people chosen randomly in the sample
  - E.g. if the person we want is not at home, we cannot simply ask the person next door
  - Why not?
- Inability to answer
  - If the question being asked is of an opinion, we can add a "don't know" option
  - If the question being asked is of a matter of fact, is there some other way of measuring? Or can the respondent try harder?

7.1 Errors of Non-observation

- Refusal to answer
  - Generally, rates of refusal have increased over recent years
  - Perhaps over worries about privacy/crime
  - A large refusal rate is a problem as it could introduce a bias
    - E.g. say most of the people who refuse to answer are elderly. Then their observations will not be included in your sample
  - We should try to find out who/what type of people refused to answer, to recognise potential biases

7.1 Errors of Non-observation

- How can we reduce refusal rates?
  - Contact respondents in advance, by letter or phone
  - Build up trust between interviewer and respondent (personalized letters, tokens of appreciation)
  - If the respondent is interested in the topic of the survey, this helps
    - But be careful: only selecting those interested in the topic could introduce biases
  - Interview length is important. Don't ask too much time of people (or don't tell them how long it will take)

7.1 Errors of Non-observation

- A high non-response rate is often a sign of a poor survey
- But not always
- The key is whether a significant section of the population is left out of the sample as a result
  - A small non-response rate that includes all the 20-30 yrs old females is a worry
  - A relatively large non-response rate that is representative of the entire population is not too bad
  - Hence the importance on gathering information on the non-responses

7.2 Errors of Observation

- We can have access to a respondent who is willing to answer our question, yet still things can go wrong
- We will say there are four types of error of observation
  - Error due to the interviewer
  - Error due to the respondent
  - Error due to the measurement instrument
  - Error due to the method of data collection

7.2 Errors of Observation

- The interviewer can have a huge influence on how the respondent answers a question
  - Most people dislike confrontation and will agree with the interviewer, if they perceive he/she has an opinion
  - Therefore, how the question is read (intonation, emphasis etc.) is very important
  - Interviewers should be friendly, without an impression of favouring one answer over any other

7.2 Errors of Observation

- The motivation of a respondent to answer a question correctly is important
  - A question may be embarrassing to answer (‘Did you view pornography in the past week?)
  - The question may be too long/boring, so the respondent answers just to hurry the interview along
- The ability of a respondent to answer a question is also important
  - Does he/she actually understand the question and the options for the answer?
  - Could they make a mistake in interpreting the units used in the question?

7.2 Errors of Observation

- People give different answers to different interviewers
- Example – Schuman and Converse (1971)
  - Black residents is Detroit were asked “Do you personally feel that you can trust most white people, some white people, or none at all?”
  - The interviewers were either white or black
  - With a white interviewer, 35% of respondents said they could trust most white people
  - With a black interviewer, 7% did

7.2 Errors of Observation

- People may say whatever they think the interviewer wants them to say

7.2 Errors of Observation

- We can categorize errors due to the respondent into the following
  - Recall bias
    - Respondent recalls information incorrectly
  - Prestige bias
    - Respondent exaggerates answer to look good
  - Intentional deception
    - Respondent deliberately lies
  - Incorrect measurement
    - Respondent did not understand the question correctly

7.2 Errors of Observation

- How many glasses of water do you drink a day?
  - Depends how big a 'glass' is…
- Are you unemployed?
  - Does it count if I'm a student and not looking for work? What if I don't want to work?
- Errors in the definition of important words in the question can cause inaccurate responses

7.2 Errors of Observation

- These are errors due to the measurement instrument
- Concepts should be defined clearly, precisely and unambiguously
  - Hold up a standard glass, then ask 'How many glasses of water do you drink a day, if you used glasses like this?'

7.2 Errors of Observation

- Interviewers, respondents and measurement instruments come together in the method of data collection
- We consider 4 methods
  - Personal interviews
  - Telephone interviews
  - Self-administered questionnaires
  - Direct observation

7.2 Errors of Observation

- Personal interviews require the interviewer to prepare questions and to record the respondent's answers
- Advantages
  - People respond better to people
  - Interviewer can correct any misunderstandings immediately
- Disadvantages
  - Interviewers can affect (perhaps unknowingly) the answer of the respondent, causing bias
  - Human error

7.2 Errors of Observation

- Telephone interviews can also be used to gather information from respondents
- Advantages
  – Cheaper than personal interviews
  – People respond better to people
- Disadvantages
  – Difficult to get a complete frame (telephone book?)
  – Must be shorter than personal interviews (people get fed up quicker)

7.2 Errors of Observation

- Self-administered questionnaires are usually posted or emailed to respondents
- Advantages
  - Cheaper than interviews
- Disadvantages
  - Not much contact with respondent, hence very high non-response rate
  - Lots of potential for bias to be introduced (accentuated when using the web)

7.2 Errors of Observation

- If a survey does not involve measuring attributes of people, direct observation may be a good method of collecting data
- E.g. estimating animal populations
- Advantages
  – Less room for bias
- Disadvantages
  – Human error

7.3 Reducing Errors

- We've known that there are a lot of potential errors in surveys
- And so far, we have only learned how to deal with sampling error
- How can we reduce all the other kinds of error?
- By sticking to a good sampling plan

7.3 Reducing Errors

- We have seen that non-response is a major source of error for surveys
- The non-response rate can be reduced by callbacks
  - A callback is another attempt to make an observation from a sampling element
- E.g. you called Mr. Smith for a phone interview but he didn't pick up. Try again the next day, at a different time of day.

7.3 Reducing Errors

- We can reduce the non-response rate by encouraging people to respond
- Rewards may encourage people
  - $
  - Supply of product
- Rewards should be offered to participants after they have been selected to be in the sample
- Offering rewards first, then selecting a sample from those who respond, risks introducing bias

7.3 Reducing Errors

- Incentives work on the same idea as rewards: people are more likely to respond to a survey if they feel they will benefit
- E.g. a letter from the Vice-Chancellor saying that this survey will help improve the quality of food in the university canteens

7.3 Reducing Errors

- The interviewer can introduce bias into results
- Hence interviewers can be trained to be better, so that they
  - Don’t lead the respondent towards an answer
  - Encourage honest responses
  - Can tell the difference between someone who doesn’t know the answer and someone who is reluctant to answer

7.3 Reducing Errors

- Like all areas of statistics, there is room for human error in surveys, particularly in recording the data (typos etc.)
- Many of these can be easily checked
  - Most people have less than 10 children
  - Proportions must be between 0 and 1
- Data checks can be performed after the results have been collected to make sure that the sample is representative (do we have 50% male, 50% female proportions in our sample?)

7.4 Question Design

- Perhaps the most important part of reducing non-sampling error is the design of the questionnaire
- As we will see, the wording of questions, the ordering of questions, the options presented as answers etc. can have a huge impact on results

7.4 Question Design

- It is possible to ask the same questions in different orders and receive different responses
- Example – McFarland (1981)
  - How interested would you say you are in religion: very interested, somewhat interested or not very interested?
  - Did you, yourself, happen to attend church in the last seven days?
- OR
  - Did you, yourself, happen to attend church in the last seven days?
  - How interested would you say you are in religion: very interested, somewhat interested or not very interested?

7.4 Question Design

- In this example, the questions are similar, but one is specific and one is general
- In McFarland (1981)
  - In the first ordering, 56% of respondents said they were very interested in religion
  - In the second ordering, 64% of respondents very interested in religion

– It is usually better to ask the more general question first, then follow with the more specific question

7.4 Question Design

- Example
  - A survey asked questions about people's happiness generally and their happiness in marriage
  - In the order 'Are you happy generally?' then 'Are you happy in your marriage?' 52% said they were very happy generally
  - In the order 'Are you happy in your marriage?' then 'Are you happy generally?' only 38% said they were very happy generally

7.4 Question Design

- The preceding questions in a questionnaire can change the frame of mind of a respondent
- Consider this sequence of questions
  - Did you watch Tottenham Hotspur F.C. win the F.A. Cup in 1991?
  - Did you know that Tottenham Hotspur F.C. beat Manchester United 6-1 earlier this season?
  - Do you think Tottenham Hotspur F.C. is a good football team?

7.4 Question Design

- With this sequence of questions
  - Did you know that Tottenham Hotspur F.C. have not won the league title since 1963?
  - Did you know that they have just lost to Arsenal, who are the worst team in the world?
  - Do you think Tottenham Hotspur F.C. is a good football team?
- Also see this clip. “Politicians use statistics rather like how drunks use lampposts – for support rather than illumination.” – Andrew Lang

7.4 Question Design

- Order of possible responses is important too
- Consider a personal interview where the interviewer gives a long list of complicated possible responses
  - The respondent is more likely to choose the last possible response, as he has forgotten what the interviewer said at the beginning

7.4 Question Design

- Conversely, consider a written questionnaire, in which the respondent is given a long list of complicated possible responses
  - The respondent is more likely to choose an early response (high up the list) because they can't be bothered to read all the options

7.4 Question Design

- How can we reduce these problems?
  - Produce questionnaires with different orderings of questions for different subsets of the sample
  - Repeat the question or use extra materials (showcards) to make sure that the question and all its possible answers are understood
  - When analyzing the data, make clear in what context the question was asked

7.4 Question Design

- A closed question is one that
  - either has a fixed number of possible, predetermined choices OR
  - Has a single numerical answer
- Closed questions are
  - easier to analyze, especially if the analysis is to be done by computer
  - Restrictive as they limit the possible response
  - Possibly influential as the possible answers may lead the respondent
  - Or may prompt the respondent to remember responses that might otherwise have been forgotten

7.4 Question Design

- An open question is one that
  - Allows the respondent to answer how they wish
- Open questions
  - Can yield more interesting and nuanced answers,
  - But they are hard to analyze
  - Because it can be difficult to quantify the response

7.4 Question Design

- Example – Schuman and Scott (1987)
  - Asked an open question: “What do you think is the most important problem facing this country today?”
  - The most common responses were: “unemployment” (17%) and “general economic problems” (17%).
  - They also asked a closed version…

7.4 Question Design

  - "Which of the following do you think is the most important problem facing the country today: the energy shortage; the quality of public schools; legalized abortion; pollution; or, if you prefer, you may name a different problem as most important."
  - The most common response was "the quality of public schools" with 32%
- A sensible plan might be to use the open question in a pre-test to determine possible fixed alternatives for the closed question in the larger sample

7.4 Question Design

- Sometimes, the respondent simply doesn’t know the answer to a question, or doesn’t have an opinion
- Such a response effectively reduces the sample size and increases the sample error
- So it is common for questionnaires not to allow such responses
- i.e. the respondent is forced to choose an answer

7.4 Question Design

- But this seems silly: how can force people to give an opinion about a subject they know nothing about?
- A sensible approach is to use screening questions to see if the respondent might have enough knowledge to answer the question properly
- If it seems that the respondent knows enough, the interviewer will ask the question without 'Don't know' as a possible response
- If it seems that the respondent does not know enough, the interviewer will skip the question

7.4 Question Design

- "Don't know" response can be included in the list of possible responses
- Generally, people can be forced into giving an answer/opinion
- But if there is an easier option, they may take that

7.4 Question Design

Without a 'Don't know' option

- Question
  - Do you feel that almost all of the people running the government are smart people, or do you think that quite a few of them do not know what they are doing?
- Responses
  - Are smart 37.0%
  - Don't know what they are doing 58.1%
  - Don't know (volunteered) 4.8%

With a 'Don't know' option

- Question
  - Do you feel that almost all of the people running the government are smart people, or do you think that quite a few of them do not know what they are doing, or do you not have an opinion on that?
- Responses
  - Are smart 28.9%
  - Don't know what they are doing 49.7%
  - No opinion 21.4%

7.4 Question Design

- Consider this question
  - ‘Is Tottenham Hotspur the worst team in England or the best team in England?’
- Notice we are only offered two extremes – there is no middle ground
- A middle-ground option is an easy option for a respondent to take

7.4 Question Design

- Questions without a middle-ground option force the respondent to think about the direction of their response
- We could include more options to allow for how strongly the respondent feels about their answer
- But generally speaking, we like to keep the number of possible responses small

7.4 Question Design

- The phrasing of a question is very important
- What is wrong with the following:
  - Do you favour the use of compulsory lectures in CUHK?

7.4 Question Design

- What is wrong with the following:
  - Do you favour the use of compulsory lectures in CUHK?
- This is yes-no question, but it is unbalanced. A more balanced form would be
  - Do you favor or oppose the use of compulsory lectures in CUHK?

7.4 Question Design

- What is wrong with the following:
  - Do you think that lectures in CUHK should be compulsory or do you think that students should be allowed to decide for themselves as adults whether to attend lectures or not?

7.4 Question Design

- What is wrong with the following:
  - Do you think that lectures in CUHK should be compulsory or do you think that students should be allowed to decide for themselves as adults whether to attend lectures or not?
- This question contains an argument against compulsory lectures. Questions should contain as little argument/counterargument as possible
  - Do you think that lectures in CUHK should be compulsory or are you opposed to this idea?

7.4 Question Design

- What's wrong with the following:
  - Do you agree that Tottenham Hotspur F.C. is by far the greatest team the world has ever seen?

7.4 Question Design

- What's wrong with the following:
  - Do you agree that Tottenham Hotspur F.C. is by far the greatest team the world has ever seen?
- This question leads the respondent into agreeing with the interviewer
- People usually do not like to contradict or conflict with other people
  - Do you think that Tottenham Hotspur F.C. is by far the greatest team the world has ever seen or not?

7.4 Question Design

- Compare the following:
  - Do you think CUHK should not allow students to miss lectures?
  - Do you think CUHK should forbid students to miss lectures?

7.4 Question Design

- Compare the following:
  - Do you think CUHK should not allow students to miss lectures?
  - Do you think CUHK should forbid students to miss lectures?
- These questions mean the same, but ‘forbid’ is much stronger word (causing stronger feelings) than ‘not allow’
- The tone of a question is very important

7.4 Question Design

- What's wrong with the following (from Lohr book)
  - "Do you agree with Bill Clinton's $50 billion bailout of Mexico?"

7.4 Question Design

- This is an example of a double-barrelled question
- It mixes the opinions of two things: Bill Clinton and the loan to Mexico
  - Disapproval of either would lead to a “no” response
- We should aim to ask about only one concept per question

7.4 Question Design

- What’s wrong with the following:
  - How much water do you drink a day?

7.4 Question Design

- What's wrong with the following:
  - How much water do you drink a day?
- Firstly, the respondent is unlikely to know in units (ml? Fluid oz?) how much water they drink
- Does the interviewer only mean water by itself? Or can tea/coffee/Coca-cola/etc. be included too?
- Questions must be specific, with all parts well-defined so that the respondent fully understands what is being asked

7.4 Question Design

- People are not as good at remembering as they think they are
  - People can struggle to remember even the simplest facts
  - People determine frequencies by decomposing, not simple counting. That is, they will establish a rate for a shorter period of time and then multiply it
    - How many times did you eat at Coffee Corner this semester?
  - People telescope events: ones they remember well seem more recent, whereas less memorable events seem to have occurred longer ago

7.4 Question Design

- We can compensate for these errors in our questionnaire design by
  - Asking questions about facts in more than one way. Even better, use direct observation
  - Decompose questions: e.g. how much water (by itself) do you drink a day? How much tea? How much beer? Etc.
  - Relate questions about events to important, memorable events to compensate for people's poor recollection of when things happened. E.g. Did you last watch Tottenham Hotspur before or after your mid-term?

7.5 Likert Scales

- You should have seen lists like these for some survey questions
  - From a CUHK Course and Teaching Evaluation form
  - From the UKOOG survey on fracking

Strongly Disagree	Disagree	Slightly Disagre	Slightly Agree	Agree	Strongly Agree
(1)	*(2)*	*(3)*	*(4)*	*(5)*	*(6)*

	1	2	3	4	5
Course quality	**Strongly Disagree**	**Disagree**	**Neutral**	**Agree**	**Strongly Agree**
1. Learning outcomes are relevant					

		Total
Unweighted base		4086
Weighted base		4086
NET: Agree		2466 60%
Strongly agree	(+2)	772 19%
Agree	(+1)	1695 41%
Neither	(0)	701 17%
Disagree	(-1)	277 7%
Strongly disagree	(-2)	235 6%
NET: Disagree		512 13%
Don't know		406 10%
lean		0.68
itandard deviation		1.08
itandard error		0.02

7.5 Likert Scales

- They are examples of Likert items and are used to gauge opinions
- *Rensis Likert* was a US psychologist who invented his method to measure attitudes in the 1930s
- His idea was that since possible responses are comparable for different questions, the responses could be assigned scores
  - "Strongly disagree (-2)", "Disagree (-1)" etc.
- These scores could then be summed (or averaged) to represent a respondent's overall opinion towards the subject of the survey
  - This sum is the Likert scale

7.5 Likert Scales

- Example – a Likert Scale about Tottenham Hotspur FC (Spurs)
  - Please select the number below that best represents how you feel about THFC

	Strongly Agree	Agree	Undecided	Strongly Disagree	Disagree
Spurs are a good football team	5	4	3	2	1
Spurs play attractive football	5	4	3	2	1
I like Spurs' uniform	5	4	3	2	1
Spurs are the best team in London	5	4	3	2	1

7.5 Likert Scales

- In that example, there are 4 Likert items in the Likert scale and a respondent's score would lie between 4 and 20.
- When designing a Likert scale, consider the two parts
  - The statements
  - The response scale
- The statements should be short, simple, clear… the same qualities we have discussed before for survey questions
  - What's wrong with this statement?
    - Spurs are poor defensively and waste money on players

7.5 Likert Scales

- The Likert response scale can take many forms, according to the question
  - Strongly agree, agree, neither agree nor disagree, disagree, strongly disagree
  - Completely agree, mostly agree, slightly agree, slightly disagree, mostly disagree, completely disagree
  - Always, very frequently, occasionally, rarely, very rarely, never
  - Very good, good, acceptable, poor, very poor
  - Extremely, very, moderately, slightly, not at all
  - Etc. etc.

7.5 Likert Scales

- How many responses should the scale have?
  - You want to offer enough choice to measure not only direction of opinion but also its strength
  - You want to make the options clear and manageable for the respondents (can they tell the difference between the 9th and 10th options in your 13 point scale?)
- In practice, either 5 or 7 possible choices is appropriate

7.5 Likert Scales

- How about 6 points?
  - Odd-numbered response scales are much more common because they are symmetrical (Strongly Agree/Strongly Disagree) with a neutral midpoint like “Undecided” or “Neither agree nor disagree”
- It is considered good practice to have a neutral midpoint
  - You don’t force respondents into giving an opinion they don’t have
  - It has been shown that even if people genuinely do not have an opinion, they are still reluctant to choose a ‘Don’t know’ option, making their actual choice more or less random

7.5 Likert Scales

- A major advantage of Likert scales is that they allow the surveyer to cover many issues and considerations of a subject with the respondent and arrive at a single number representing their attitude toward it
  - You cannot do this with a single question, hence the need for several Likert items to make up one Likert scale

7.5 Likert Scales

- Example
  - the British Social Attitudes (BSA) survey contains this Likert scale (called the left-right for measuring the political leftness or rightness of the respondent
  - The response scale is “agree strongly”, “agree”, “neither agree nor disagree”, “disagree”, “disagree strongly”

- *Government should redistribute income from the better off to those who are less well off*
- *Big business benefits owners at the expense of workers*
- *Ordinary working people do not get their fair share of the nation’s wealth*
- *There is one law for the rich and one for the poor*
- *Management will always try to get the better of employees if it gets the chance*

7.5 Likert Scales

- In BSA's left-right scale,
  - "agree strongly" to "disagree strongly" are given the scores to 1 to 5;
  - the average over the items in the scale gives the left-right index for an individual
- – They also have
  - the libertarian-authoritarian scale
  - The welfarism scale
- – You can read the latest BSA report in their website.

7.5 Likert Scales

- How many Likert items should a Likert scale have?
  – The more the merrier, but of course you will be restricted by time
- This means the items must be chosen carefully, so they cover the key aspects of the subject and measure the overall attitude of the respondent
  – Much, much research goes into what the items should be

7.6 Real Life Survey: the HK HES

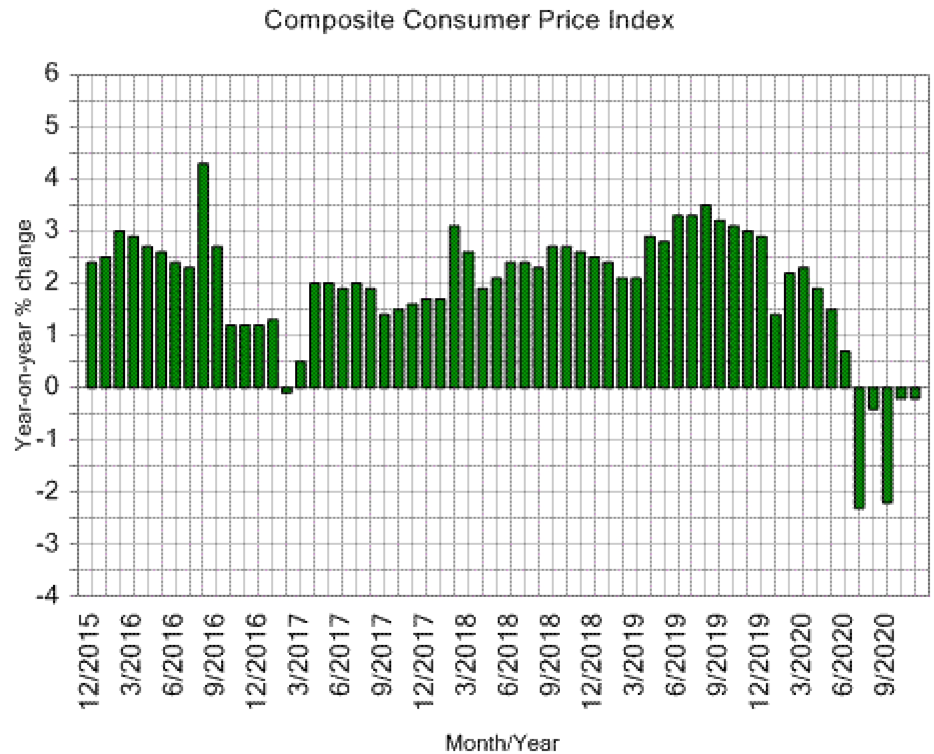
- Many of the response variables we consider can be measured directly
  - Height, weight, sugar content of oranges etc.
- Our questionnaires allow for more abstract concepts
  - Are you happy in your marriage?
  - On a scale of 1-10, how much do you love Tottenham Hotspurs F.C.?
- Consider these questions
  - Are Hong Kong students becoming smarter?
  - Which is richer: Hong Kong or Singapore?
  - Do you think Hong Kong becoming more expensive?

7.6 Real Life Survey: the HK HES

- To answer these questions properly, we need to analyse a vast amount of data
  - Can we turn this complicated question into one number, such that, as the number changes, we learn about the question at hand?
- Such a number is called an index – they are constructed for summarizing a complicated situation
- Examples
  - We summarize the economy of a country by its Gross Domestic Product (GDP)
  - Hang Seng summarizes the performance of the stock market in Hong Kong by its Hang Seng Index
  - To summarize the cost of living in Hong Kong, the HK Government calculates the Consumer Price Index (CPI)

7.6 Real Life Survey: the HK HES

- Let’s focus on the CPI
- It is produced by the Census and Statistics Department
- They produce thousands of statistics about Hong Kong, as well as charts like this



(from https://www.censtatd.gov.hk)

7.6 Real Life Survey: the HK HES

- You may know the CPI as a measure of inflation, or a measure of how much cheaper/more expensive things have become over a year
- The CPI can be interpreted these ways. More accurately, it is a "measure to reflect changes in the price level of consumer goods and services generally purchased by households"
- It is calculated using surveys
  – The Household Expenditure Survey (HES)
  – A Monthly Retail Price Survey

7.6 Real Life Survey: the HK HES

- The HES is conducted every five years, to collect information on what HK households spend their money on
- The 2014/15 HES
  - Began with a pretest of 200 households in May to July of 2014
  - Took one year, from Oct. 2014 to Sept. 2015, to cover the seasonal variation in consumer habits

7.6 Real Life Survey: the HK HES

- Their frame had two parts – the Register of Quarters (RQ) and the Register of Segments (RS)
  - RQ is a database of all address of permanent quarters In built-up areas. Each unit of quarters is identified by a unique address (e.g. street name, building name, floor number, flat number)
  - RS is a database of segments in non-built-up areas. Segments may contain 8-15 quarters each, and these may not have clear addresses, hence using the segments as the sampling unit

7.6 Real Life Survey: the HK HES

- The design used is stratified random sampling with proportional allocation
  - The strata were chosen as the homogeneous groups that were likely to have similar spending patterns
    - E.g. type of housing (private, subsidized, public, temporary), geographical area (The Peak v Tai Po)
  - With these considerations, 11 strata were chosen to form homogeneous groups. 10 from RQ, 1 from RS.
- Next, within each of the 11 strata, systematic sampling of quarters/segments was applied. The same sampling fraction (i.e. the $1/k$) was used in the 10 RQ strata, a smaller one was used for the RS stratum

7.6 Real Life Survey: the HK HES

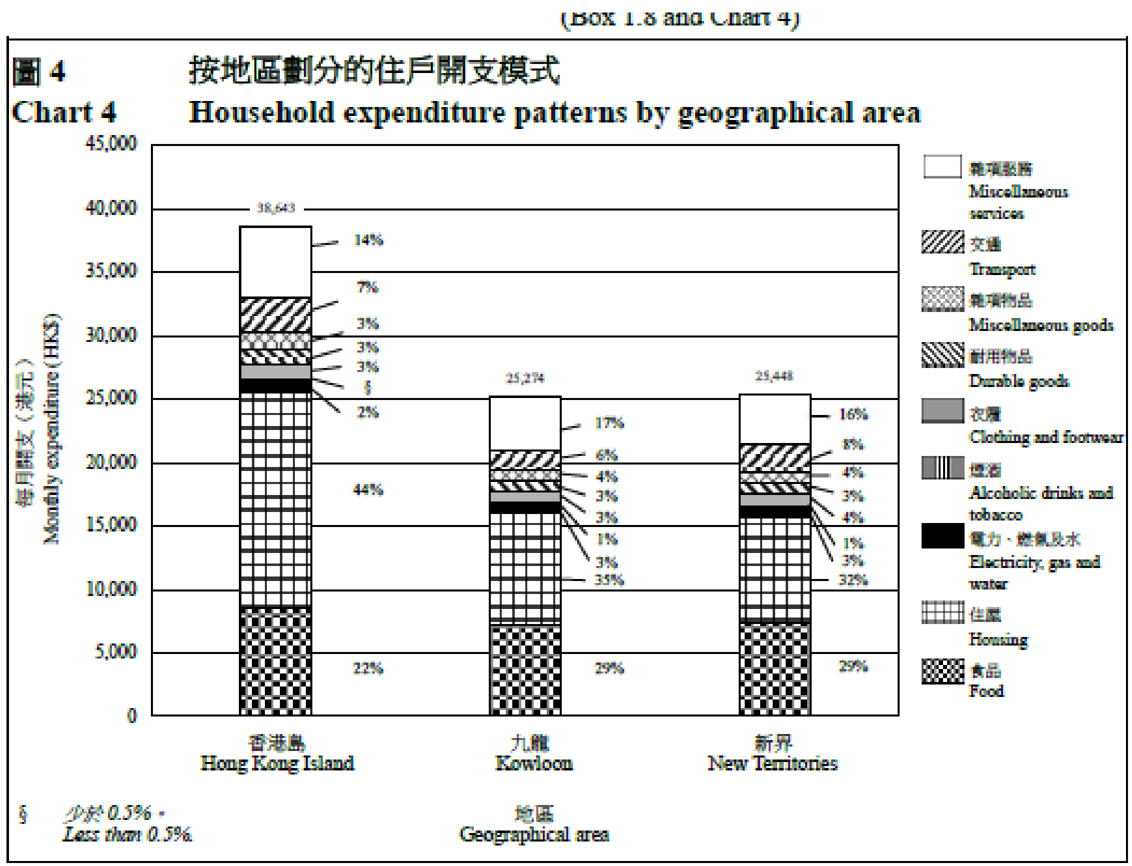
- The sample size was 12,000 quarters
  - This was to allow for non-response and unsuitable cases (quarters not used for residential purpose, or being vacant, etc.)
  - They hoped to have 6,800 responses
  - A high non-response rate was likely. Not only would some sampled quarters be ineligible (see above), but the survey required a household to keep a diary of what they bought for 2 weeks
  - Many households would provide some information, but would drop out before the 2 weeks had finished or they would refuse to do the full survey, but would complete a simple questionnaire

7.6 Real Life Survey: the HK HES

- To deal with cases like these
  - Weighting adjustments were used, with weights (approximately) equal to the inverse of the class response rate
  - Elsewhere, imputation was used for missing data (we will learn about imputation later)
- 12,547 quarters were selected from the sampling frame
  - 842 were found to be ineligible (vacant, demolished, not used for residential purpose)
  - From the 11,705 remaining quarters, 11,787 households were covered, as some quarters accommodated more than one household
  - Of these 11,787 households, 9,416 were found suitable for the survey (e.g. they would not be absent for the survey reference period)
  - Of these 9,416 households, 6,812 participated in the survey
  - Of the remaining 2,604 that did not participate in the survey, 749 could not be contacted, 1,536 refused, 319 were drop-outs

7.6 Real Life Survey: the HK HES

- This produces a lot of data about what households in Hong Kong spend their money on, and how much they spend



(From the 2014/15 HES)

7.6 Real Life Survey: the HK HES

- How does this impact the CPI
  - Households spend more on some items (rice) and less on others (toothpaste)
  - Thus price movements on different items will have different impacts on the amount the household spends
  - So a weighting system is used to denote the relative importance of things households buy
- Essentially, the HES tells the Census and Statistics department what goods and services households buy and how important they are relative to each other

7.6 Real Life Survey: the HK HES

- Thus a representative basket of 984 goods and services is created.
- These are divided into 94 groups
- And these are divided further into 241 sub-groups

表 5.1 綜合消費物價指數中的商品／服務組別指數（2014年10月至2015年9月 = 100）（續）
Table 5.1 Composite CPI at commodity/service group level (Oct. 2014 - Sep. 2015 = 100) (Cont'd)

組別編號 Group no.	商品／服務類別及組別 Section and group of commodity/service	權 數 Weight	每年平均數 Annual average 2017	2018	2019	2019年指數與2018年指數比較 Index for 2019 compared with that for 2018 (%)	在總指數變動率中所佔的比率^ Contribution to rate of change in all-item index^ (%)	在類別指數變動率中所佔的比率^ Contribution to rate of change in section index^ (%)
20.	蛋 Eggs	0.12	100.9	103.6	104.7	+ 1.1	+ 0.0 *	+ 0.1
21.	食油 Edible oils	0.14	94.7	95.1	94.3	- 0.9	- 0.0 *	- 0.1
22.	汽水 Carbonated drinks	0.05	99.9	103.9	109.0	+ 5.0	+ 0.1	+ 0.2
23.	其他不含酒精飲品 Other non-alcoholic beverages	0.26	101.1	103.2	104.8	+ 1.6	+ 0.1	+ 0.3
24.	糖 Sugar	0.02	104.4	109.6	112.2	+ 2.3	+ 0.0 *	+ 0.0 *
25.	糖果 Confectionery	0.13	99.2	97.6	97.7	+ 0.1	+ 0.0 *	+ 0.0 *
26.	調味品及配料 Flavourings and additives	0.11	99.9	102.0	104.3	+ 2.3	+ 0.1	+ 0.2
27.	其他食品 Food, others	0.92	107.1	112.1	115.0	+ 2.6	+ 0.9	+ 1.8
	住屋 HOUSING	34.29	106.8	109.5	113.3	+ 3.5	+ 42.2	+ 100.0
28.	租金（連差餉及地租） Rent, including rates and government rent	31.86	106.6	109.1	112.8	+ 3.4	+ 38.4	+ 90.7

(From "Annual Report on the Consumer Price Index 2019"*, by Census & Statistics Department)

7.6 Real Life Survey: the HK HES

- The basket of goods and their weights are updated every 5 years, after the HES
  - Items can enter (smartphones) or leave (VHS) the basket of goods
- Every month, a Monthly Retail Price Survey is conduced by the Census & Statistics Department
  - This finds out the current prices of items in the basket of goods, from different sources, throughout HK
  - Data is collected mainly by visits (about 10,000 field visits), but also by telephone (about 1,000) are made to about 4,000 retail outlets
  - In total, around 47,000 price quotations are collected this way
  - Other sources include the HK Housing Authority (for rents) and utilities (for electricity, post, transport etc.)

7.6 Real Life Survey: the HK HES

- After the data has been collected, a price index is calculated for each consumer good or service
  - This index reflects the change in price of the item over the time period consider
- E.g. Say the price of 6 rolls of toilet paper on Feb 28th 2019 was \$30. On March 31st we find they cost \$31.
  - The price index for 6 toilet rolls would be $100 \times \frac{31}{30} = 103.3$
- The CPI is then calculated by multiply the price index for an item by its weight, then summing over all the items in the basket of goods

7.6 Real Life Survey: the HK HES

- Example
  - You are simple person who consumes lunch, t-shirts and DVDs every month. Nothing else.
  - Usually, you consume spend 80% of your expenditure on lunch, 15% on t-shirts and 5% on DVDs
  - At the end of February 2015, a standard lunch cost $20, a standard t-shirt cost $40 and an average DVD cost $100
  - At the end of March 2015, a standard lunch cost $21, a standard t-shirt cost $35 and an average DVD cost $100
  - Thus the item indices for your basket of goods would be: $100 \times \frac{21}{20} = 105$ for lunch; $100 \times \frac{35}{40} = 87.5$ for t-shirts; $100 \times \frac{100}{100} = 100$ for DVDs
  - Your most recent monthly CPI would then be $80\% \times 105 + 15\% \times 87.5 + 5\% \times 100 = 102.125$

7.7 Interpreting Results

- You have conducted your survey, calculated estimates and their errors for the parameters you are interested in
  - Your final step is to draw some conclusions
  - But what should they be?
  - In general, it is best to be cautious, especially when it comes to statements of cause and effect
- Confounding variables are factors that
  - Have not been considered by researchers, but affect the suspected causes and the effect

7.7 Interpreting Results

- Example – Berkeley Sex Bias (Bickel, Hammel and O'Connell, (1975))
  - In 1973, 8442 men applied for admission to graduate studies at UC Berkeley
    - 3738 were accepted
  - The same year, 4321 women applied for admission to graduate studies at UC Berkeley
    - 1494 were accepted
- Is this evidence of sex discrimination by the Graduate Division at UC Berkeley?

7.7 Interpreting Results

- What could be a confounding factor?
- From the paper”
  - “The effect may be clarified by means of an analogy. Picture a fishnet with two different mesh sizes. A school of fish, all of identical size, swim toward the net and seek to pass. The female fish all try to get through the small mesh, while the male fish all try to get through the large mesh. On the other side of the net all the fish are male.”

7.7 Interpreting Results

- Or the following hypothetical example from the paper
  - Imagine two hypothetical departments, the Department of Machismatics and the Department of Social Warfare
  - 400 men and 200 women apply to the Department of Machismatics
    - They are admitted in exactly equal proportions: 200 men and 100 women are accepted
  - 150 men and 450 women apply to the Department of Social Warfare
    - They are admitted in exactly equal proportions: 50 men and 150 women are accepted

7.7 Interpreting Results

- Thus $\frac{250}{550} = 45.5\%$ of men were accepted
- But $\frac{250}{650} = 38.5\%$ of women were accepted
- This looks like sex bias, but it can't be
  - In each department, men and women were accepted at exactly equal proportions.
- So what has happened?

7.7 Interpreting Results

- The difference comes from
  – the differing preference of departments for males and females
    - $\frac{400}{550} = 72.7\%$ of men applied to Dept. Machismatics, but only $\frac{200}{650} = 30.8\%$ of women applied to the same department
  – the departments' differing acceptance rates
    - 50% acceptance rate for Dept. Machismatics, 33% acceptance rate for Dept. Social Warfare

7.7 Interpreting Results

- Here's the data from the 6 largest departments (there were 85 (!) departments in Berkeley)
- Do you see any trend here?
- What happens when you aggregate the groups?

Dept.	# Male Applicants	% Admitted	# Female Applicants	% Admitted
A	825	62%	108	82%
B	560	63%	25	68%
C	325	37%	593	34%
D	417	33%	375	35%
E	191	28%	393	24%
F	373	6%	341	7%

7.7 Interpreting Results

- The Berkeley data is an example of Simpson’s Paradox
  - A trend that appears in groups of data can disappear or even reverse when the groups are combined
- Lesson
  - We must be careful when aggregating data
  - We must make sure that the data satisfies the necessary assumptions for the statistical techniques we use

7.7 Interpreting Results

- A (fictitious) example
  - You sample people who are colour-blind and people who are not
  - Of the people who are colour-blind, you find that 25% smoke
  - Of the people who are not colour-blind, 15% smoke
  - Your colleague suggests that the study strongly suggests that smoking increases the risk of colour-blindness
  - What do you say?

7.7 Interpreting Results

- The major point is that goal of this course was to learn how to estimate parameters well
- We have not learned how to infer cause and effect

- So be careful with how you interpret your results
  - They might suggest an association between variables, but necessarily a cause-effect relationship
- Also be careful who is interpreting the results
  - They might not be entirely objective…

7.8 Planning a Survey

- Having learned several survey designs, several survey errors and how to design a questionnaire, we are now in a position where we are able to describe a good plan for a survey
- What should be in your plan?

7.8 Planning a Survey

1. Statement of objectives
   – Decide what you want to do
   – Choose reasonable goals
   – Simpler the better
   – Stick to them throughout your survey

7.8 Planning a Survey

2. Target population
   - What is your population?
   - Define precisely what your population is
   - Choose a population for which sample selection is possible (remember you will have to select a sample from it)

7.8 Planning a Survey

3. The frame
   - Most sample designs require a frame (or frames)
   - Find and choose frames that cover as much of the population as possible
   - Use multiple frames if you want, e.g. for the population of a city, use a list of city blocks (one frame) and a list of residents within blocks (another frame)

7.8 Planning a Survey

4. Sample design
   - What sample design will you use?
   - And why?
   - Take practical considerations into account
   - What do you need to achieve your goal?
   - Determining sample size is crucial
   - Make sure your sample design enables you to produce something worthwhile

7.8 Planning a Survey

5. Method of measurement
   – How will you conduct your survey?
     - Personal interview? Telephone interview? Mailed/online questionnaire? Direct observations?
6. Measurement instrument
   – How will you obtain measurement?
   – If by questionnaire, design your questionnaire well

7.8 Planning a Survey

7. Selection and training of fieldworkers
   - Someone has to collect data
   - If interviews (telephone or personal) are to be used, training may be required
   - Fieldworkers must be told how to perform the survey, how to record measurements

7.8 Planning a Survey

8. The pretest
   - We have seen several examples of when a pretest can be extremely useful
     - Questionnaire design
     - Estimating parameters required for determining sample size
   - A good opportunity to check whether your design works practically
   - Allows you to make changes before the ‘main’ survey

7.8 Planning a Survey

9. Organization of fieldwork
   – Needs to be planned
   – Who, where and when
10. Organization of data management
   – You will be collecting lots of data. How will you manage it?
   – A major concern for large surveys

7.8 Planning a Survey

11. Data analysis
    - Once you have collected your data, what will you do with it?
    - Be clear what you will do before you collect your data
    - Strongly related to your sample design and your objective